Auxiliary input

File: aux_input.kicad_sch

Driver supply

File: supply_12v.kicad_sch

Logic and sensors

File: supply_logic.kicad_sch

Bus and brake

File: bus_brake.kicad_sch

U half bridge

File: bridge_u.kicad_sch

V half bridge

File: bridge_v.kicad_sch

W half bridge

File: bridge_w.kicad_sch

Current sensing

File: current_sense.kicad_sch

母线电压 (仅 V_DC)

File: voltage_sense.kicad_sch

Analog protection

File: comparators.kicad_sch

Hardware latch

File: fault_logic.kicad_sch

Hardware PWM gates

File: control_interface.kicad_sch

Base interlock

File: base_interlock.kicad_sch

Base LEDs

File: base_leds.kicad_sch

GPIO command coupling

File: mcu_coupling.kicad_sch

ADC and status returns

File: core_signal_protection.kicad_sch

母线电流 (15 A保护待决)

File: bus_current.kicad_sch

64-pin core sockets

File: power_interface.kicad_sch

叠板 inverter 在上：母座 J101/J102 在 B 面；B 面整面 GND（大量缝合过孔接 In1/In4），
散热铜在顶面和内层；顶面预留散热片和机械固定，不挡电解电容和焊线；无风扇，无铜条；2 oz 铜；
J4 MR60 放上边缘，距 C20，功率级左移 8–10 mm 再下移 2–3 mm；母线分流开尔文走线；
验收 5 Arms/12–60 V/20 kHz，环境 ≤30 °C；布局见 #4

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

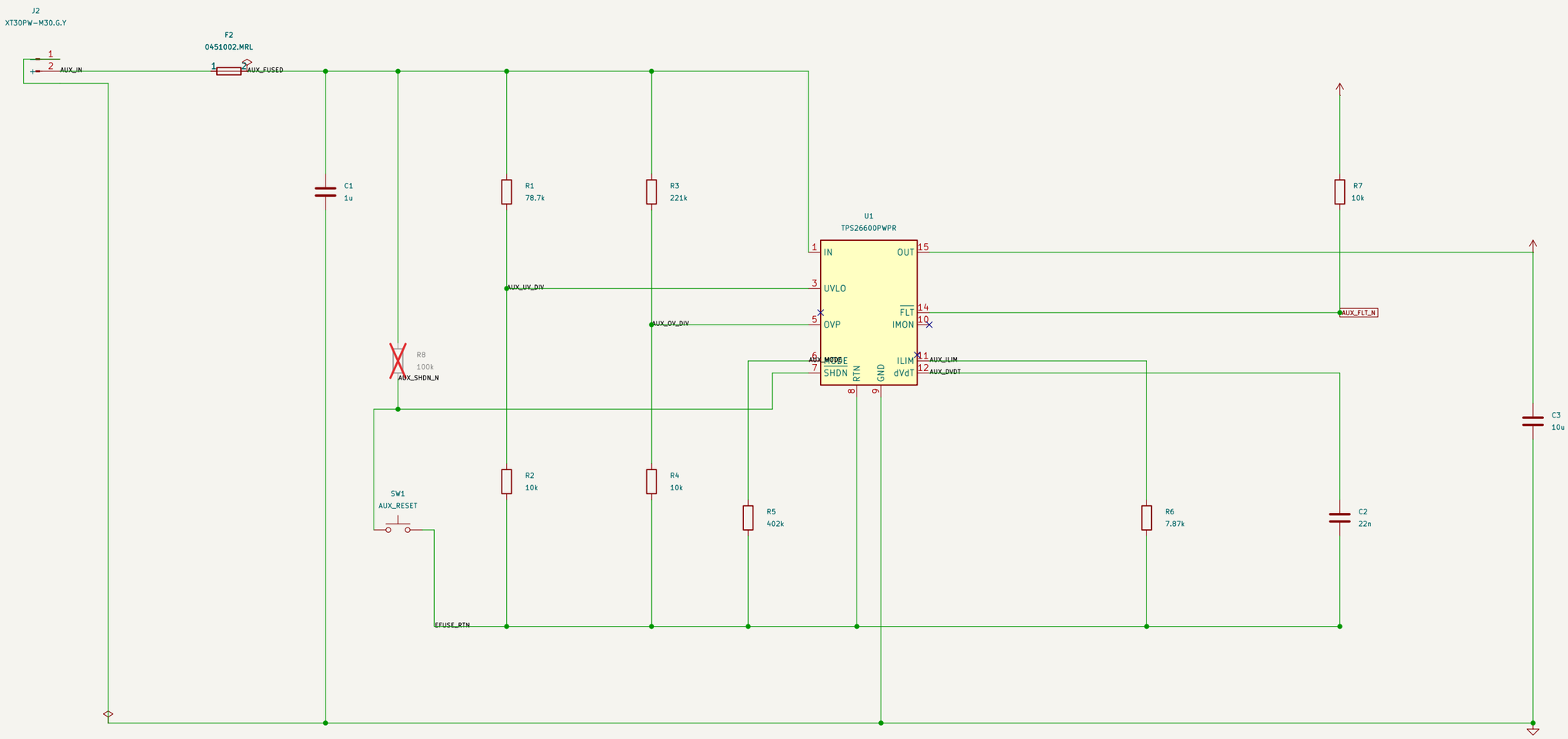
power_system

Sheet: /
File: inverter.kicad_sch

Title: Inverter base with GPIO minimum-system core

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 1/19

AUXILIARY INPUT / eFUSE / 12–24 V



RTN / exposed pad is the eFuse reference island. It is NOT the board GND pin.
MODE=402k selects latch-off. AUX_RESET resets the auxiliary path; no MCU startup dependency.
UV / OV / current-limit values remain unchanged; dynamics and surge coordination still require review.

J2:1=-/GND:2=+ 辅助输入（按共享库）:XT30PW-M30.G.Y。

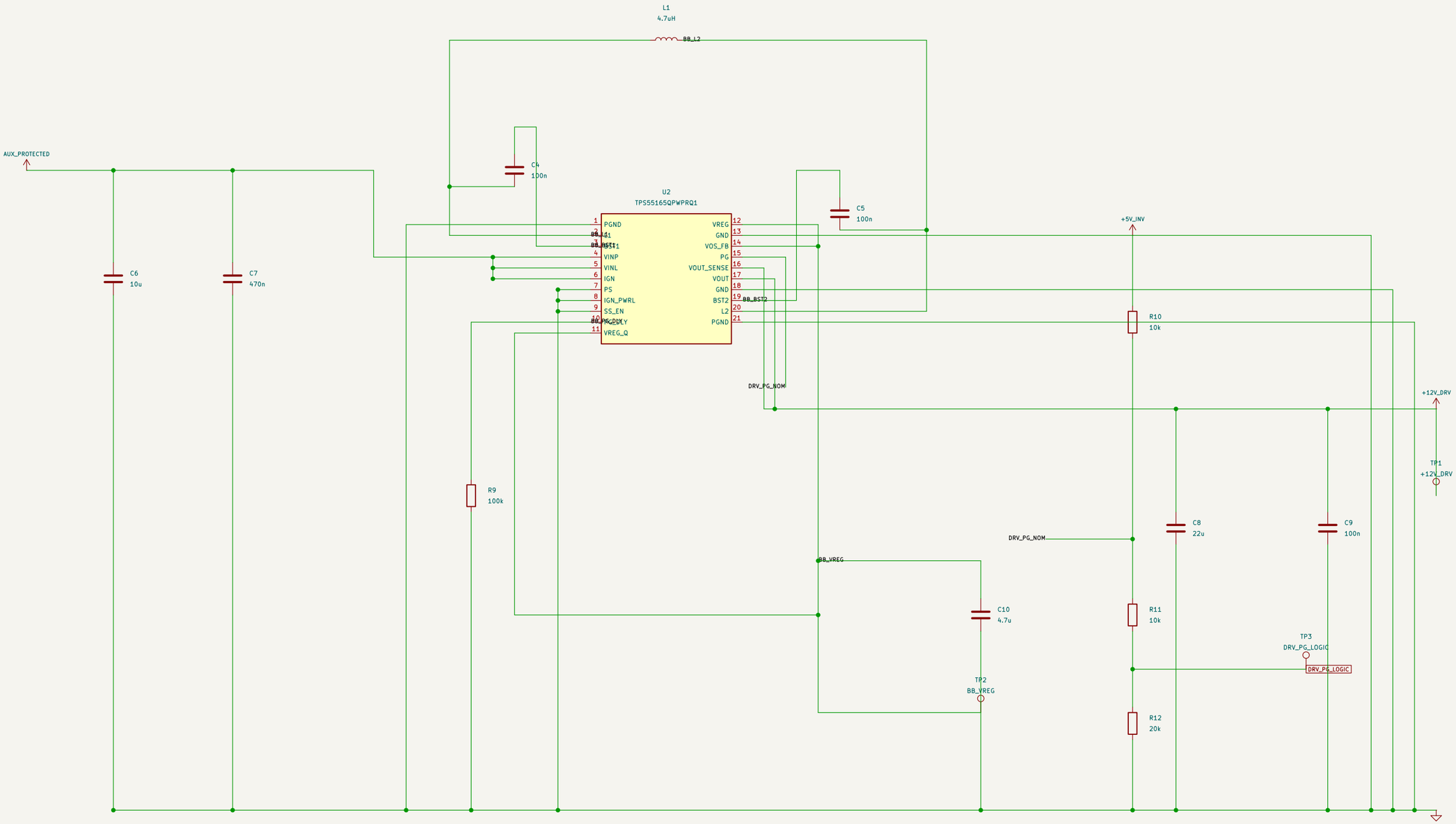
INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

Sheet: /Auxiliary input/
File: aux_input.kicad_sch

Title: Inverter – 12–24 V auxiliary input protection

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 2/19

12 V DRIVER RAIL / TPS55165 BUCK-BOOST



VREG_Q and VOS_FB return to VREG; VREG is not a load rail. VOUT_SENSE is tied to the 12 V output.
Input/output/bootstrap wiring is shown directly. PG high is about 2.5 V with the fitted pullup/divider.
Inductor model, startup, worst-case gate voltage and thermal capability remain review items.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

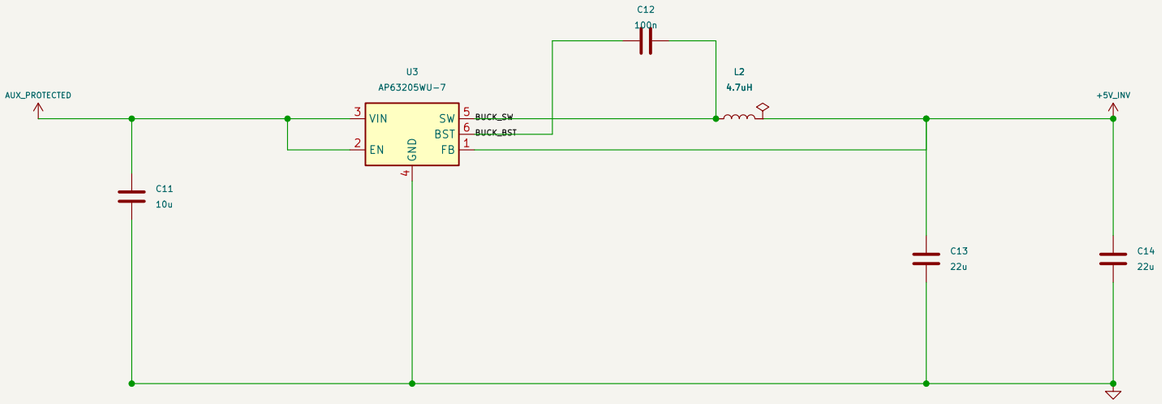
power_system

Sheet: /Driver supply/
File: supply_12v.kicad_sch

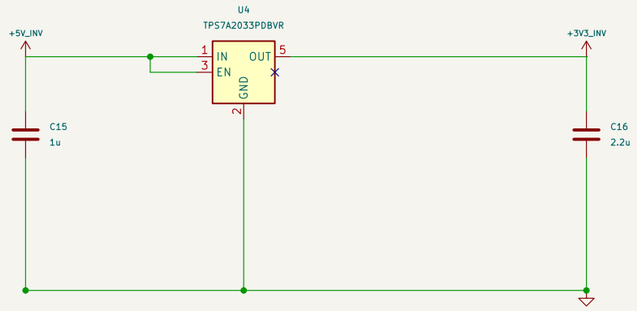
Title: Inverter - regulated 12 V driver supply

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 3/19

5 V BUCK / AP63205

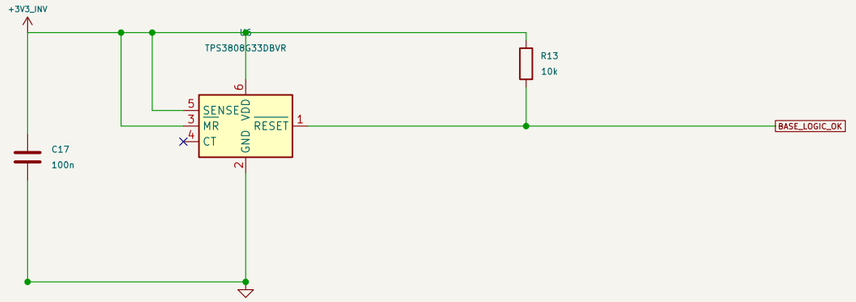


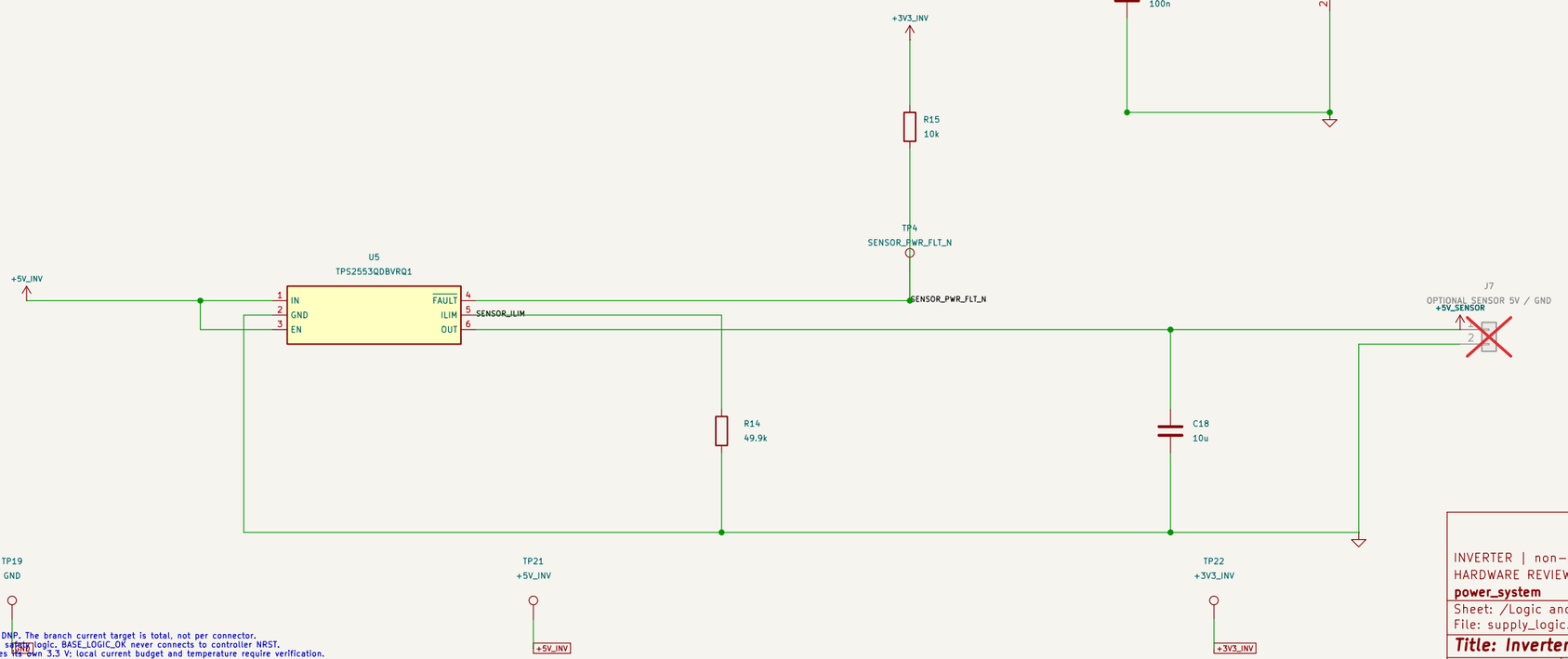
3.3 V BASE LOGIC / ANALOG SUPPLY



PROTECTED OPTIONAL SENSOR POWER / TOTAL TARGET 0.5 A

3.3 V SUPERVISOR / LOCAL BASE PERMISSION





J7 / external power links default DNP. The branch current target is total, not per connector.
U4 supplies only base analog and safety logic. BASE_LOGIC_OK never connects to controller NRST.
The removable MCU board regulates its own 3.3 V; local current budget and temperature require verification.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

power_system

Sheet: /Logic and sensors/
File: supply_logic.kicad_sch

Title: Inverter — 5 V, 3.3 V and sensor power

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 4/19

母线输入 / 理想二极管 / 母线侧储能：桥侧高频电容见 bus_current。

J3
XT60PW-M.G.Y

F1
0476008.MRSN

Q9
BSC110N15NS5ATMA1

IDEAL_GATE

GND

DC_FUSED

VBUS

C19
220u / 100V

C20
220u / 100V

TP20
VBUS

D2
SMCJ60CA / [illegible]ATE DNP

U32
LTC4359ISB#PBF

IN
GATE
SOURCE
OUT
SHDN
NC
VSS
NC

IDEAL_VSS

D212
DDZ9699T

R215
1k

C213
2.2u / 100V X7R

D213
SMAT70A-13-F

D214
SMAJ24A

J4
MR60PW-M.G.B

U_PHASE
V_PHASE
W_PHASE

U32 S8:1 GATE / 2 SOURCE / 3 NC / 4 IN / 5 SHDN / 6 VSS / 7 NC / 8 OUT。
SHDN 接 IN 常使能：VSS 按 ADI 图4经 R215=1k 回到 GND；C213 跨 OUT/VSS。
D212 碰源齐纳 12 V：D213/D214 共阳极，分别相位输入正/负瞬态（原厂推荐拓扑）。

J4：1=U、2=V、3=W；上边缘布局见 #4。
线端 MR60-F（C3040681）不上板。J5 焊线孔保持。

ONE BRAKE BRANCH / EXTERNAL RESISTOR / CONTROLLER HEADER COMMAND

J5
BRAKE POWER: BR+ / BR_SW

BR_SW

Q7
BSC110N15NS5

C23
1u

C24
100n

U29
UCC27517DBVR

IN+
IN-
VDD
GND
OUT

BRAKE_IN

BR_DRV

R17
15

BR_GATE

R16
10k

R18
10k

J5: BR+ = VBUS, BR_SW = MOS drain. External resistor goes across these pins; unused port stays OPEN.
Brake is independent of the bridge ENABLE/fault gates. The base-side input pull-down defaults a missing controller command LOW.

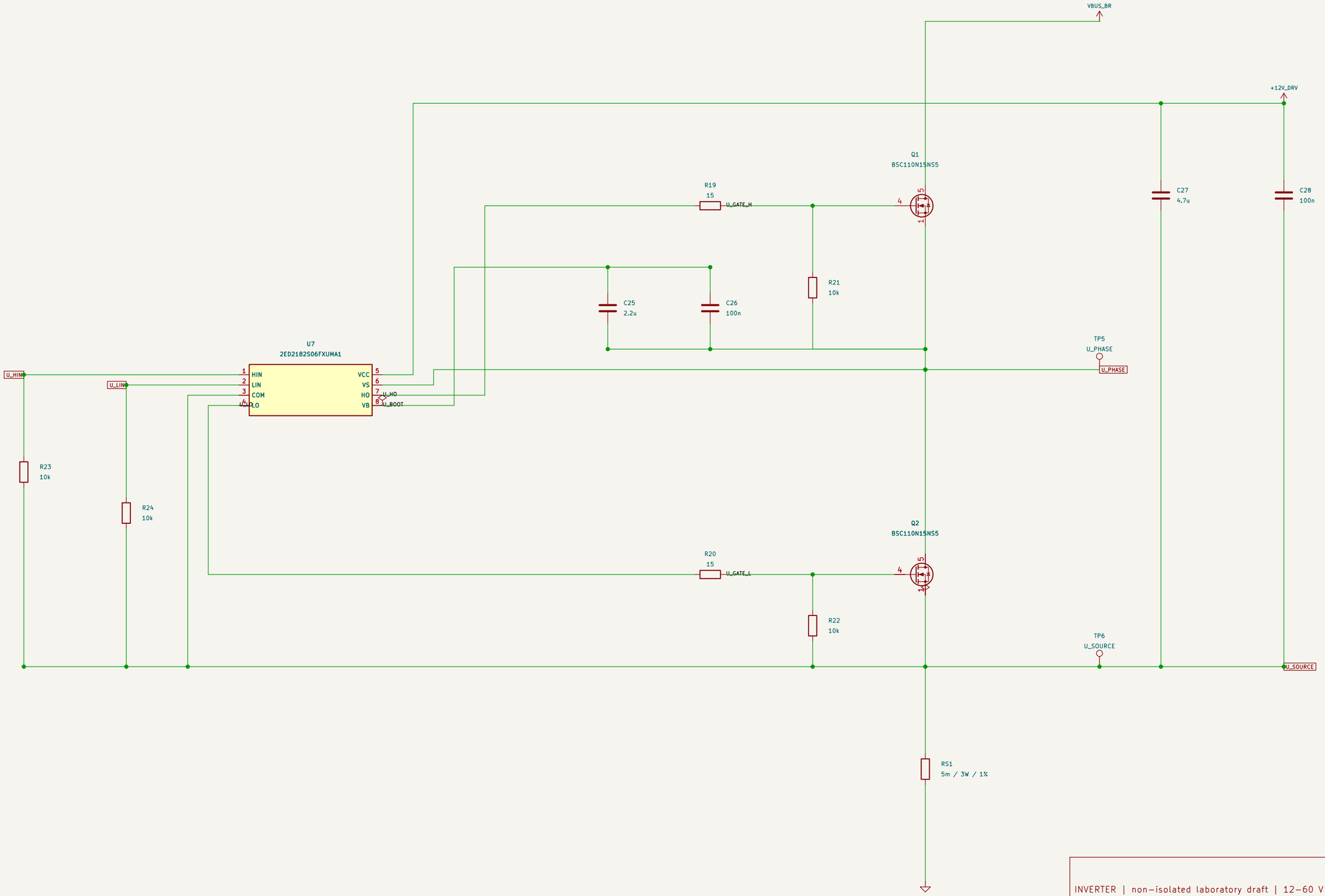
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Bus and brake/
File: bus_brake.kicad_sch

Title: Inverter — DC bus input, bulk capacitors and brake

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 5/19

U HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

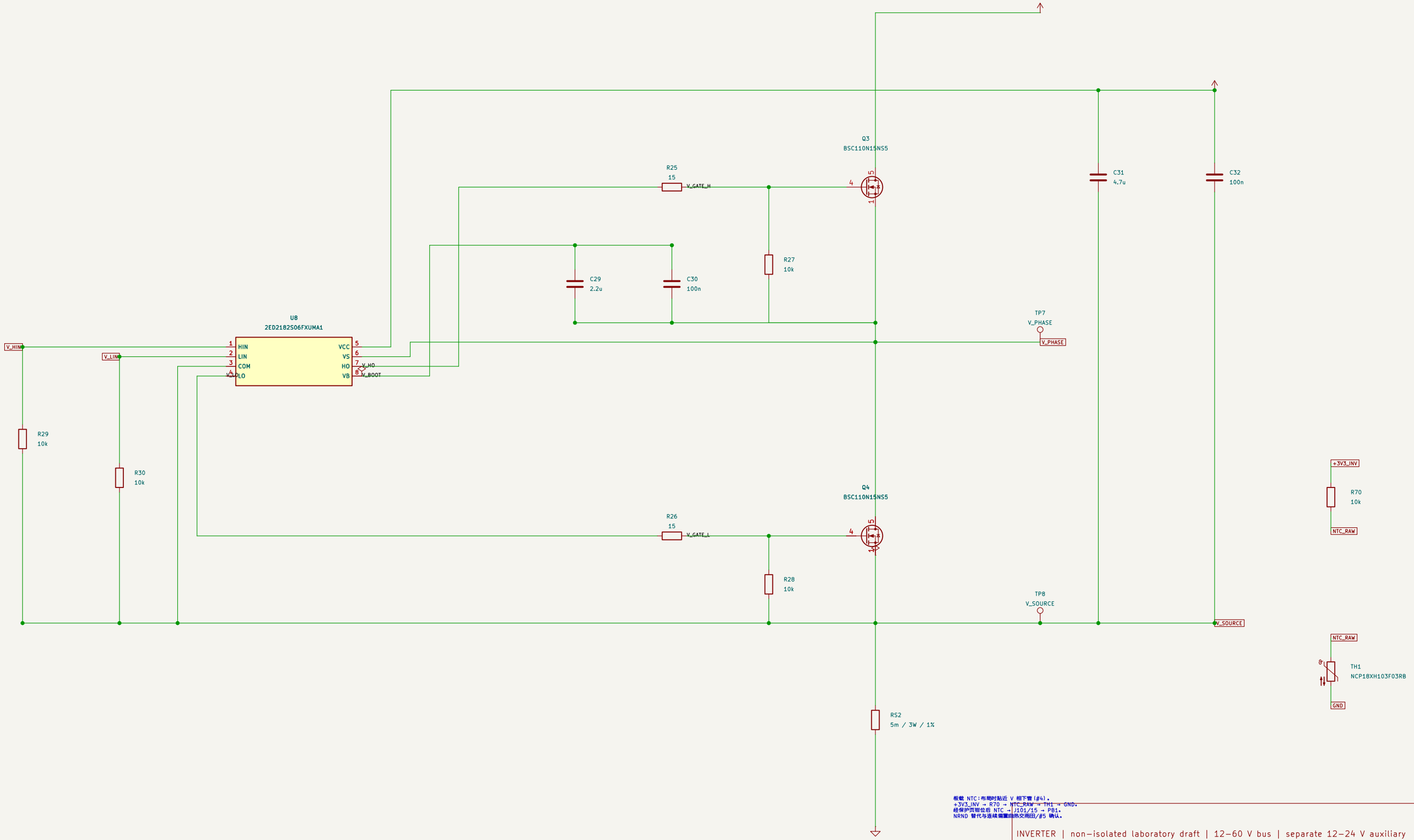
power_system

Sheet: /U half bridge/
File: bridge_u.kicad_sch

Title: Inverter - U half bridge and low-side shunt

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 6/19

V HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



板载 NTC：布局时贴近 V 相下管（#4）。
+3V3_INV → R70 → NTC_RAW → TH1 → GND。
经保护页钳位后 NTC → J101/15 → PB1。
NRND 替代与连续偏置回路交叉由田/#5 确认。

COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

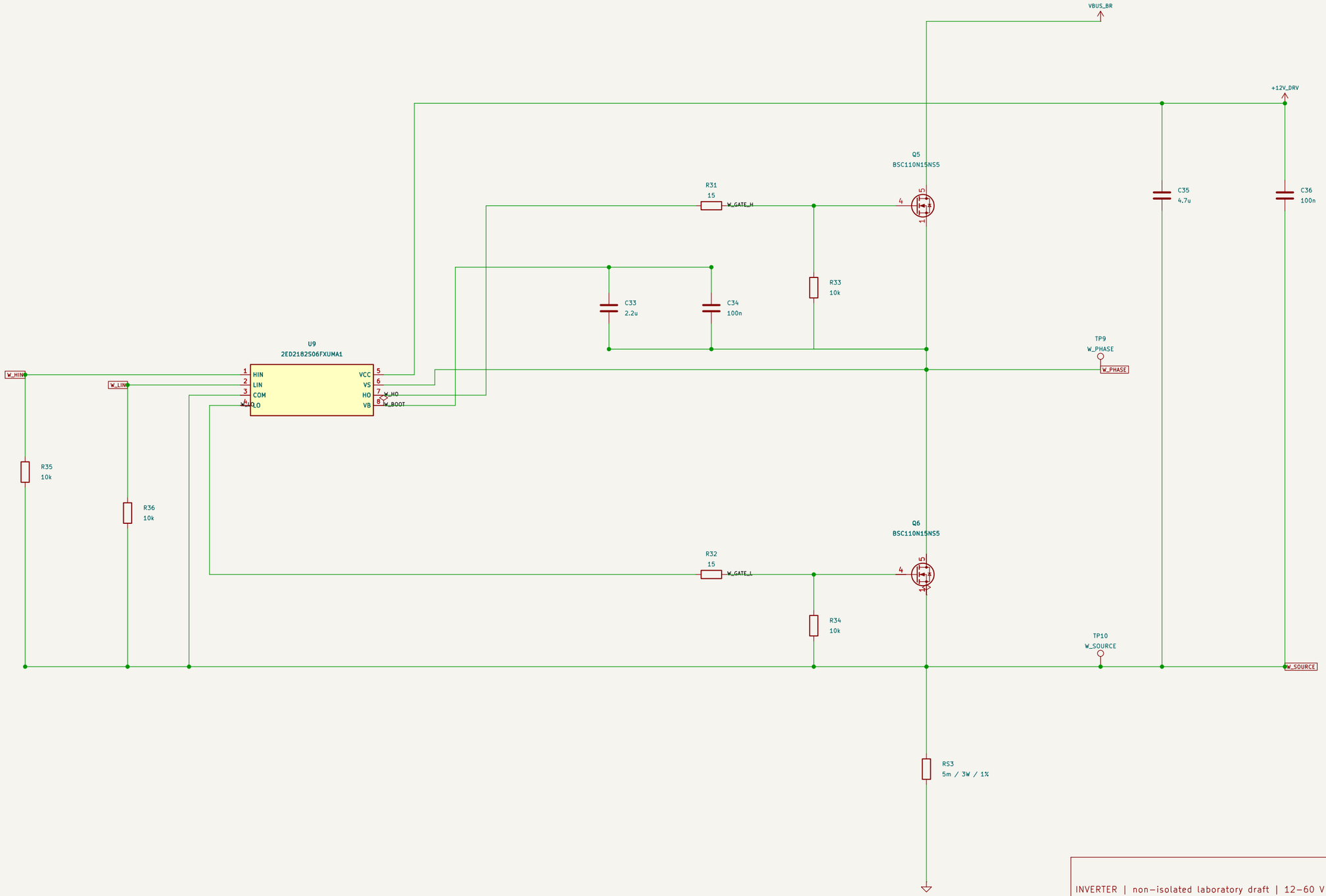
power_system

Sheet: /V half bridge/
File: bridge_v.kicad_sch

Title: Inverter — V half bridge and low-side shunt

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 7/19

W HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

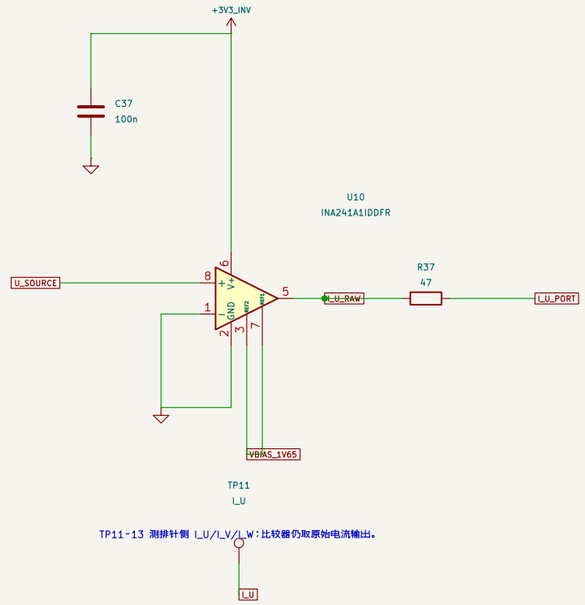
power_system

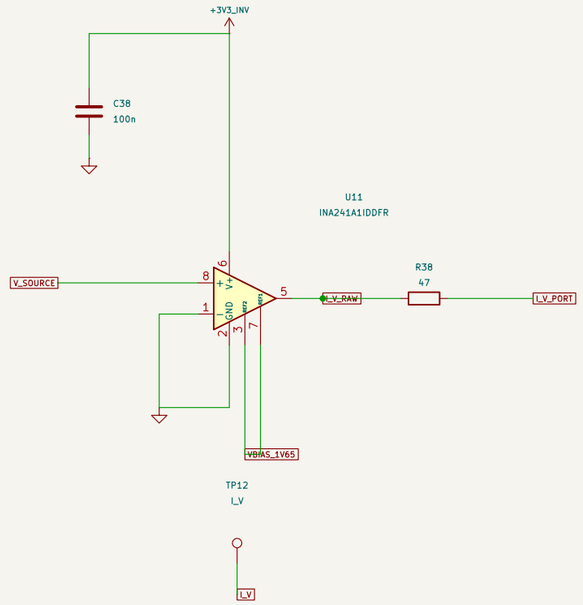
Sheet: /W half bridge/
File: bridge_w.kicad_sch

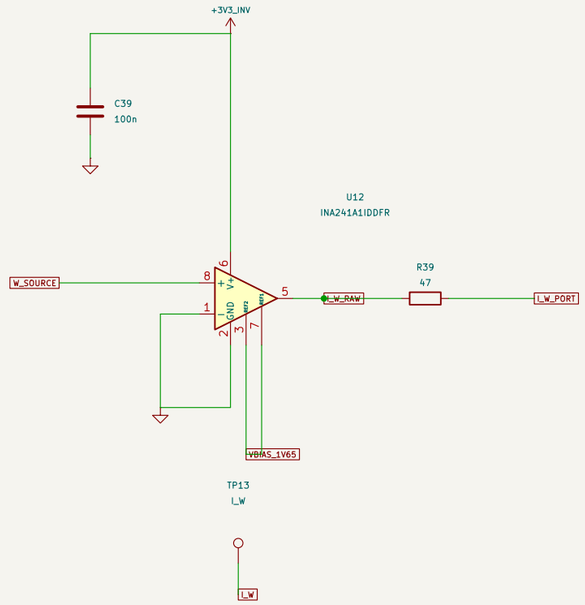
Title: Inverter - W half bridge and low-side shunt

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 8/19

THREE LOW-SIDE KELVIN CURRENT CHANNELS



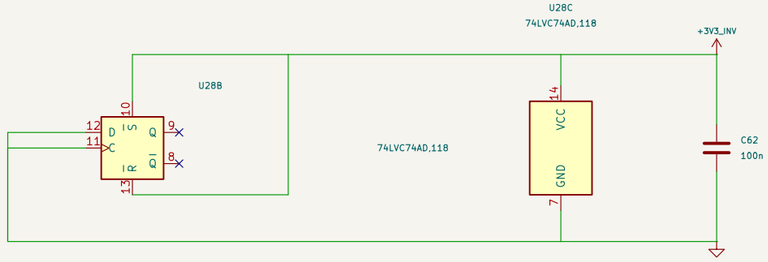




TP11-13 测排针侧 I_U/I_V/I_W：比较器仍取原始电流输出。

BUFFERED BIAS AND CURRENT WINDOW REFERENCES

SPARE U28B / FIXED INPUTS

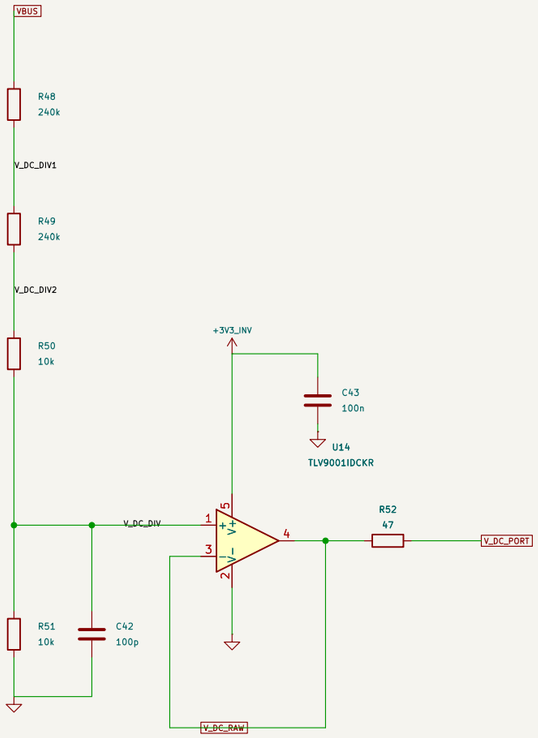


VOUT = 1.65 V + 50 mV/A·I 对应约 ±12 A（1.04995/2.25005 V）。
保留相分流 5 mΩ：低端分流的开关状态盲区由新增高端母线检测补充。

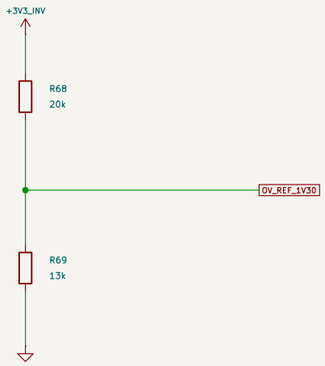
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system
Sheet: /Current sensing/
File: current_sense.kicad_sch
Title: Inverter — three current channels and references

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 9/19

仅保留 V_DC 母线采样及硬件过压参考；三相电压前端已删除。



240k + 240k + 10k / 10k → 50:1；65 V 对应 1.30 V；3.3 V 理想满量程 165 V。
原分压值保持；板载 NTC 见 bridge_v。



20k / 13k → OV_REF_1V30 = 1.30 V；母线过压名义 65 V，保持原值。

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

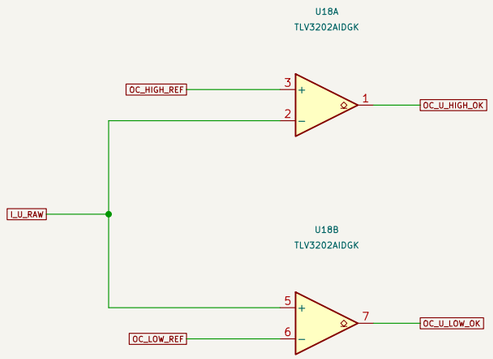
power_system

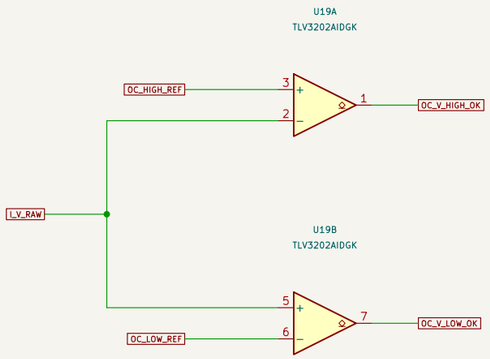
Sheet: /母线电压（仅 V_DC）/
File: voltage_sense.kicad_sch

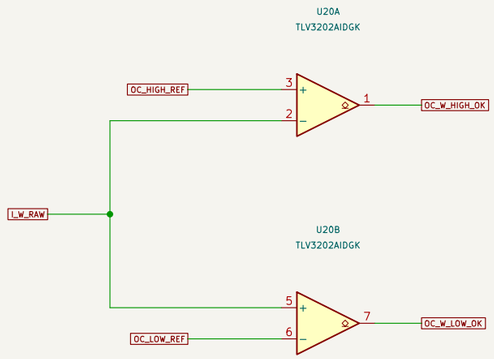
Title: Inverter — voltage and NTC sensing

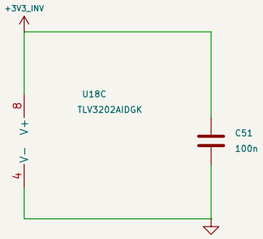
Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 10/19

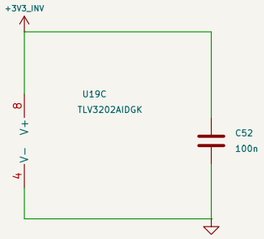
HARDWARE WINDOWS / EACH OUTPUT IS HEALTHY HIGH

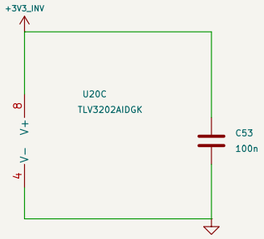




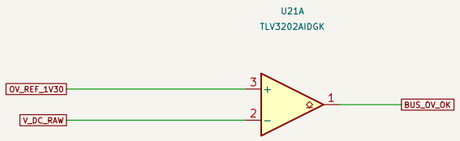


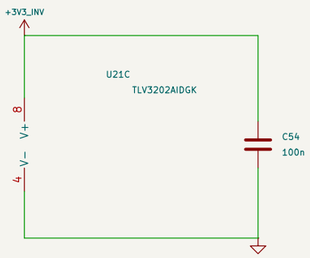


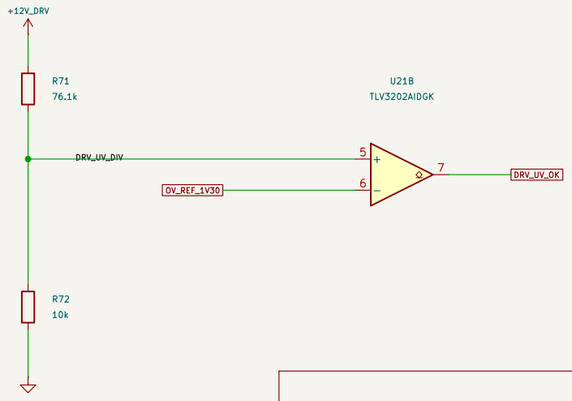




BUS OVERVOLTAGE / DRIVER RAIL UNDERVOLTAGE







健康为高：相电流越界、母线约 >65 V 或驱动电源约 <11.193 V 时对应输出拉低。
各推挽输出独立进入与门，不并联。TLV3202 内置迟滞典型 1.2 mV（约 0.024 A），无新增外部迟滞；首故障锁存。

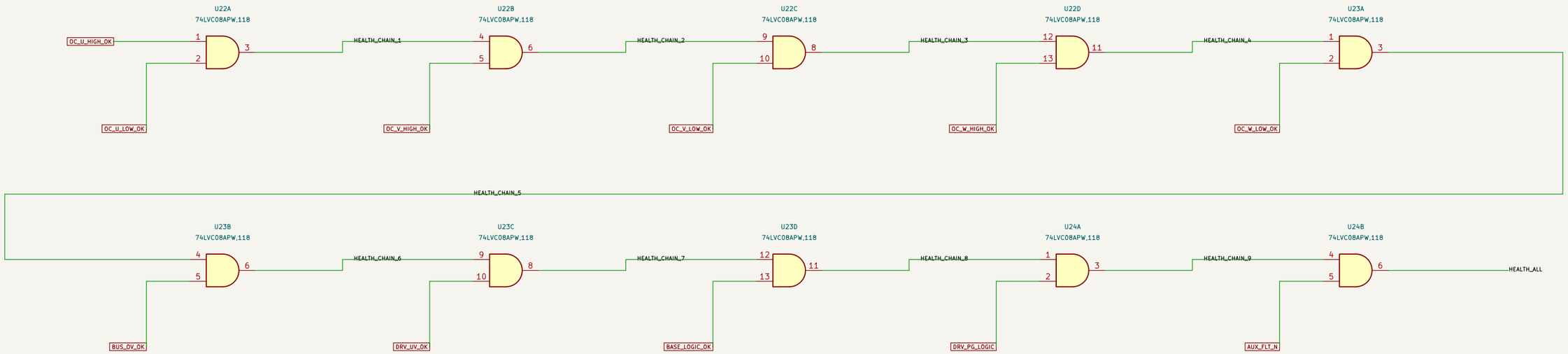
INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Analog protection/
File: comparators.kicad_sch

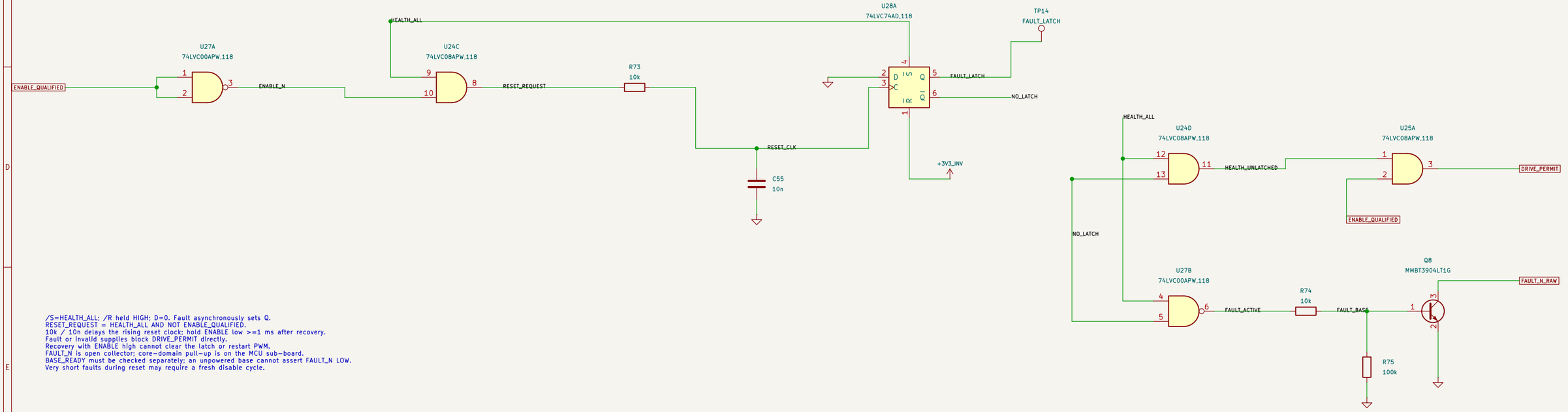
Title: Inverter — current, bus voltage and driver rail comparators

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 11/19

HARDWARE HEALTH CHAIN / EVERY INPUT MUST REMAIN HIGH



FAULT MEMORY / DELIBERATE DISABLE–TO–RESET / PERMISSION



/S=HEALTH_ALL; /R held HIGH; D=0. Fault asynchronously sets Q.
RESET_REQUEST = HEALTH_ALL AND NOT ENABLE_QUALIFIED.
10k / 10n delays the rising reset clock: hold ENABLE low >=1 ms after recovery.
Fault or invalid supplies block DRIVE_PERMIT directly.
Recovery with ENABLE high cannot clear the latch or restart PWM.
FAULT_N is open collector; core–domain pull–up is on the MCU sub–board.
BASE_READY must be checked separately; an unpowered base cannot assert FAULT_N LOW.
Very short faults during reset may require a fresh disable cycle.

INVERTER | non–isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

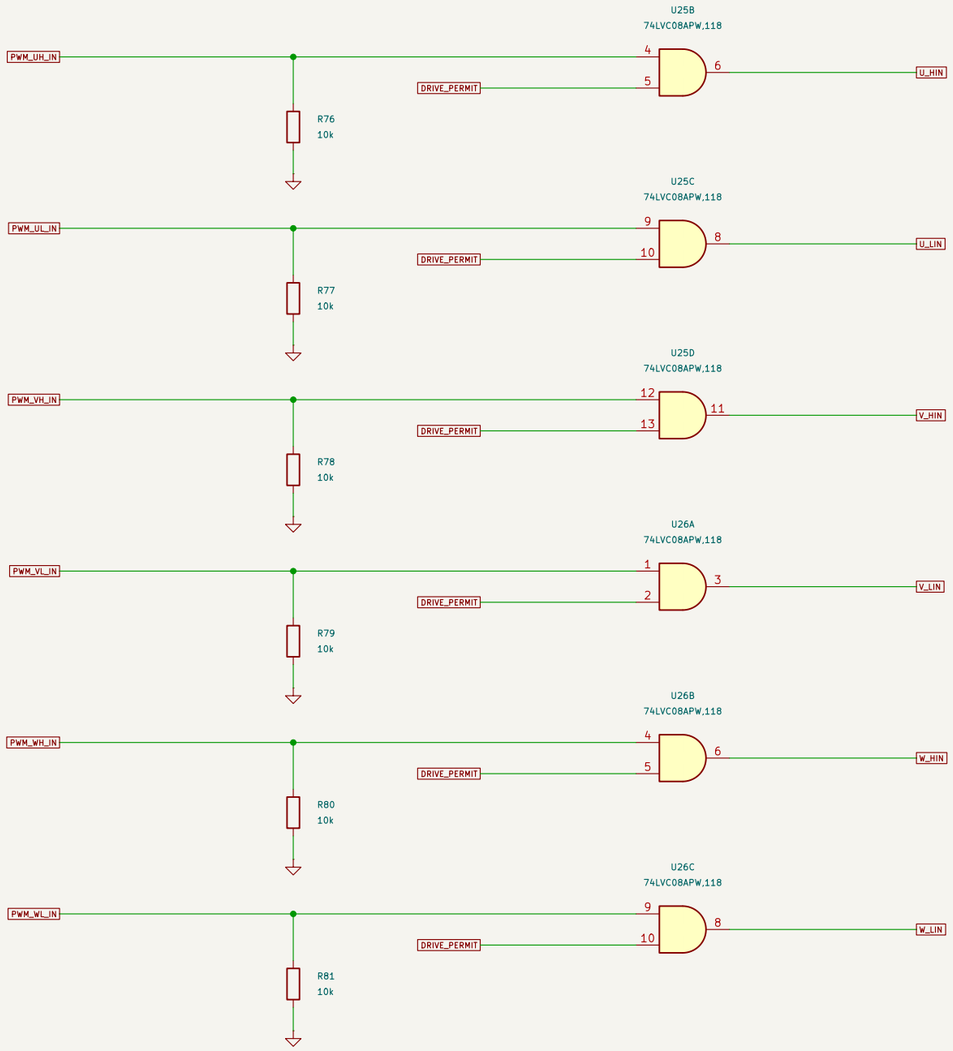
power_system

Sheet: /Hardware latch/
File: fault_logic.kicad_sch

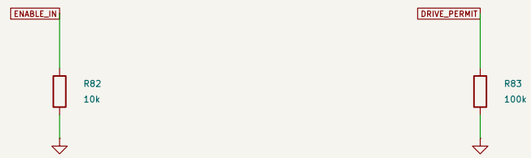
Title: Inverter – fault aggregation and reset latch

Size: A3	Date: 2026–11–12	**Rev: E–1–issue3–draft**
KiCad E.D.A. 10.0.7		Id: 12/19

HARDWARE GATING / SIX PWM COMMANDS

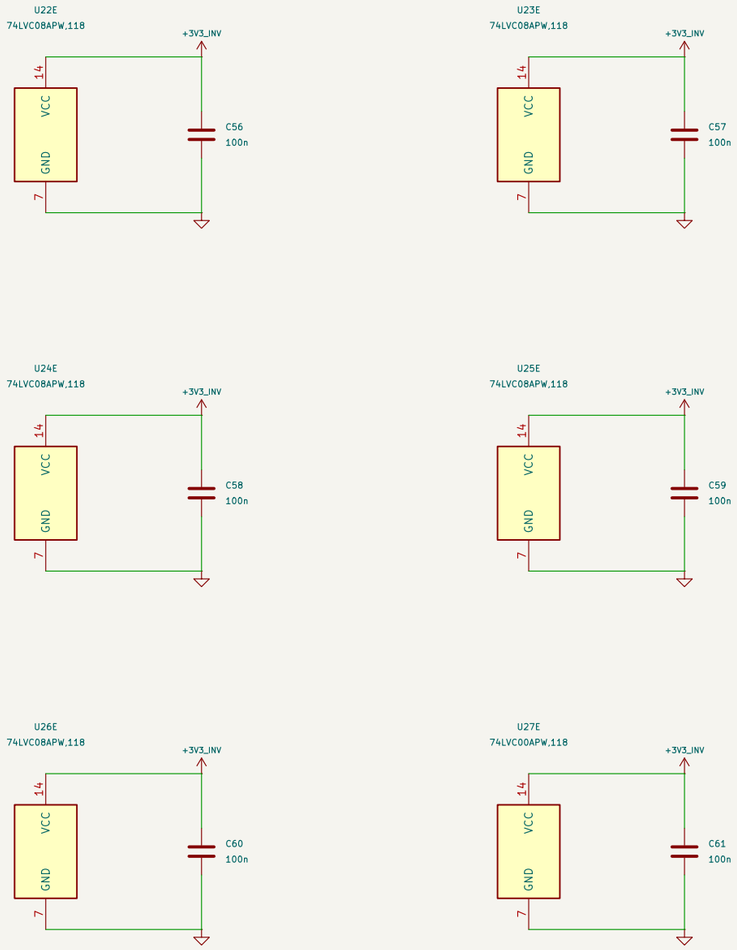


DEFAULT DISABLE / NO PERMISSION



R76–R81: hold all six commands LOW while the controller is disconnected or Hi-Z.
R82: default disabled. R83: default no permission.
Logic levels remain 3.3 V; hardware AND gates operate independently of firmware.

LOGIC POWER / LOCAL DECOUPLING



C56–C61: 100 nF local bypass at each logic package.

U26D / U27C / U27D are used on the Base interlock page.
R76–R81 default PWM LOW; R82 raw ENABLE default LOW.
No MCU on this board; CTRL_READY must qualify ENABLE before any latch reset or DRIVE_PERMIT.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

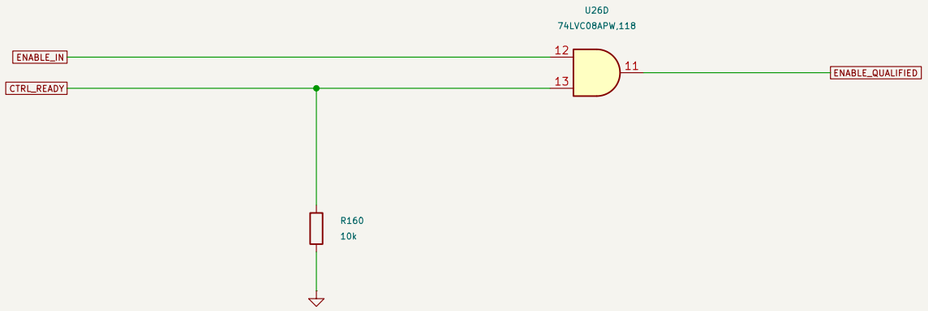
Sheet: /Hardware PWM gates/
File: control_interface.kicad_sch

Title: Inverter – local six PWM hardware gates

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 13/19

REMOVABLE CONTROLLER / DEFAULT DISABLE

共地、3.3 V 逻辑。CTRL_READY = controller U105 (RUN_ARM ∧ CORE_RESET_OK)。
功率板保留 PWM 默认下拉及硬件许可；controller 缺失或复位时不依赖固件关闭桥。



R160 = 10k. Unplugged, controller reset or lost controller power => ENABLE_QUALIFIED LOW.
Only ENABLE_QUALIFIED enters the latch reset and DRIVE_PERMIT logic. Raw ENABLE_IN keeps its original local R82 pull-down.

BASE SUPPLY PRESENCE / INDEPENDENT OF OPEN-COLLECTOR FAULT



BASE_READY = BASE_LOGIC_OK AND DRV_PG_LOGIC. The MCU sub-board provides its receiver pull-down.
FAULT_N remains Q8 open collector with a pull-up only in the MCU 3.3 V domain. A high FAULT_N alone does not prove base power.
U26/U27 package power and 100 nF bypass stay on Hardware PWM gates. No added logic package.

Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base interlock/
File: base_interlock.kicad_sch

Title: Power base — controller readiness and local supply interlock

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 14/19

LOCAL POWER INDICATOR / NO MCU LED SIGNALS ACROSS HEADER



D3: base 3.3 V present. This lamp alone does not indicate driver power or protection permission.
D4..D7 and R131..R134 belong to the removable MCU sub-board: RUN / FAULT / COMM / AUX.
The power base has no MCU-run or communication LED nets on its board-to-board header.

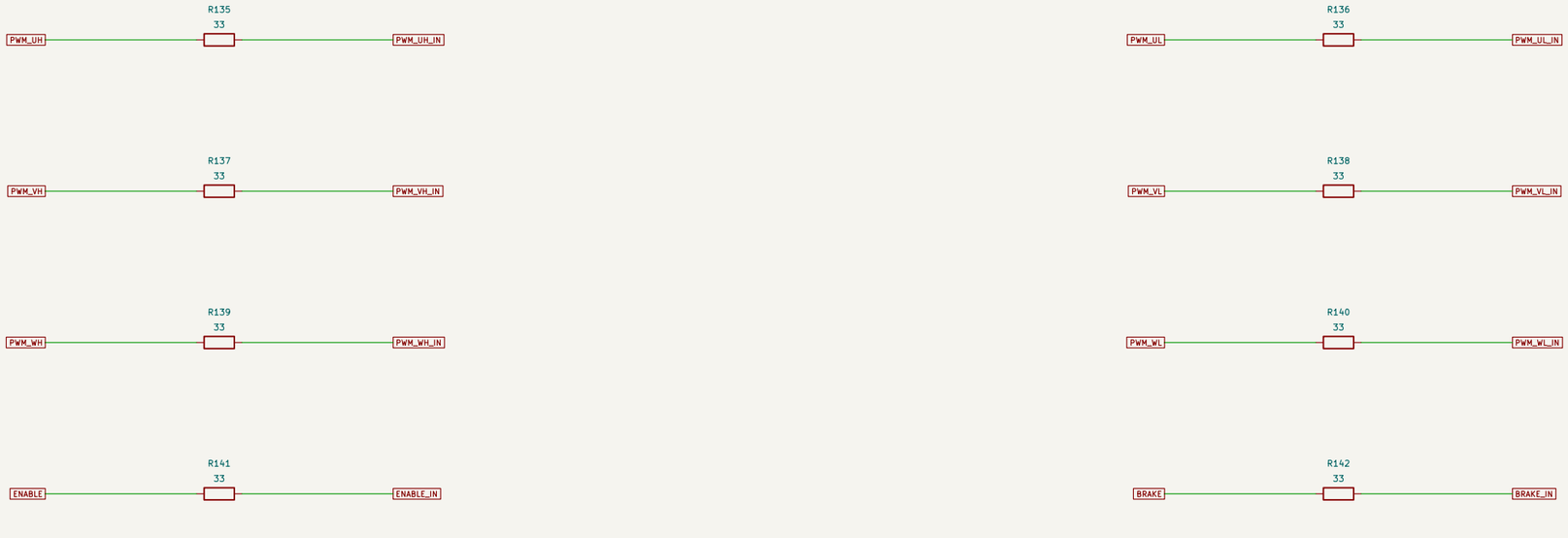
Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base LEDs/
File: base_leds.kicad_sch

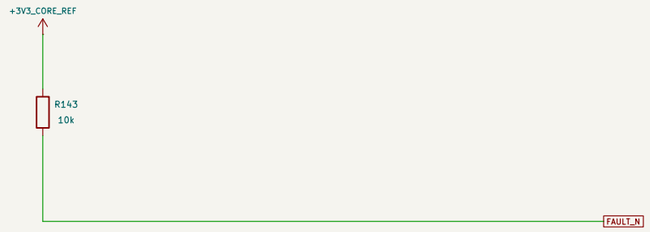
Title: Power base – local power indication

Size: A4	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 15/19

CONTROLLER COMMAND SERIES RESISTORS / BASEBOARD SIGNALS



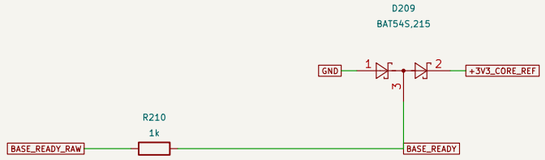
FAULT INPUT / LOCAL PULL-UP AND TIMER BKIN



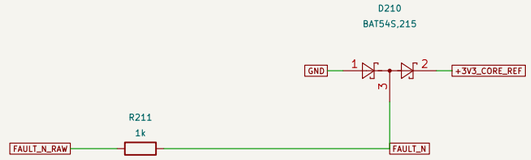
FAULT_N 经背位后到 J102/20 → PB12/BKIN；上拉只用 +3V3_CORE_REF。
六路 PWM、ENABLE 与 BRAKE 保留板内串阻；BOOT/NRST 不从本板引出。

CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system
Sheet: /GPIO command coupling/
File: mcu_coupling.kicad_sch
Title: Baseboard — GPIO command coupling and fault pull-up
Size: A3 | Date: 2026-11-12 | **Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7 | Id: 16/19

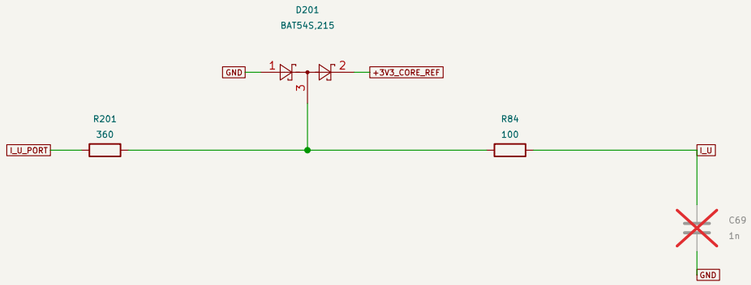
模拟接口：串阻 + BAT54S；末端 RC 在 controller；调试电容默认 DNP。



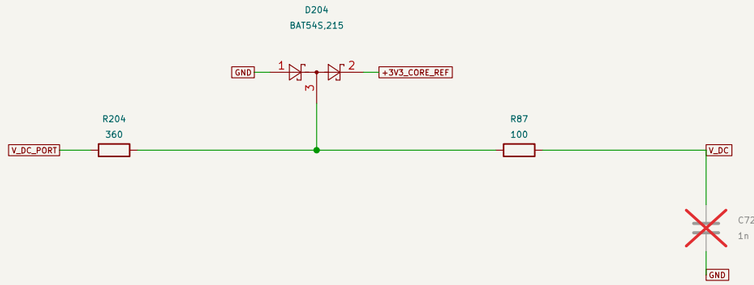




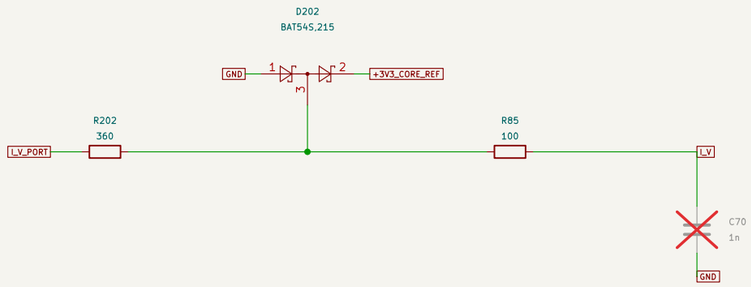
I_U_PORT → I_U



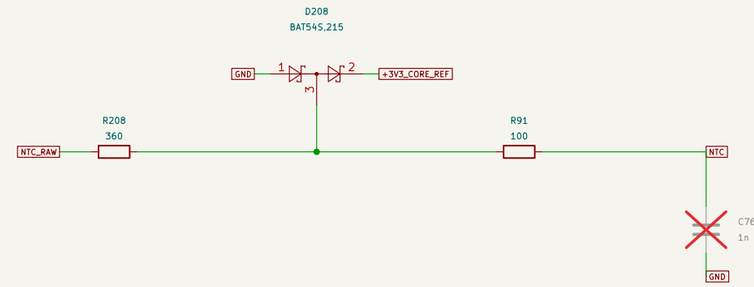
V_DC_PORT → V_DC



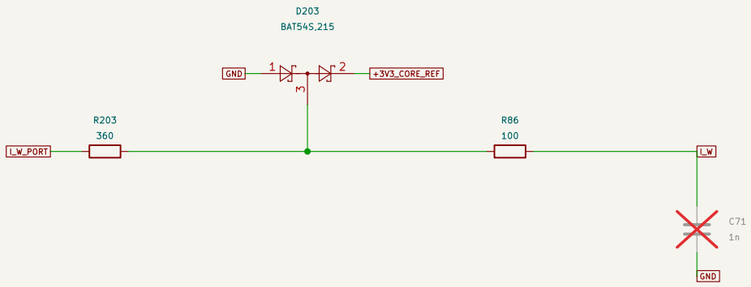
I_V_PORT → I_V



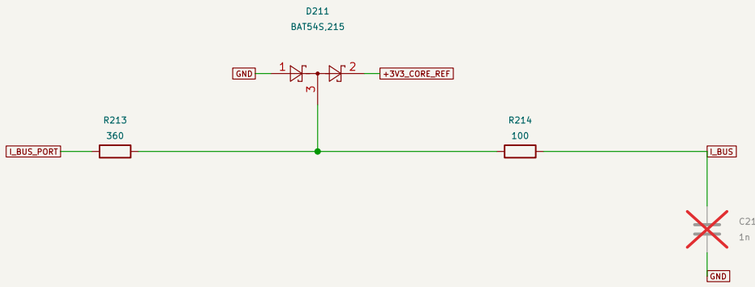
NTC_RAW → NTC



I_W_PORT → I_W



I_BUS_PORT → I_BUS



上拉位只接 +3V3_CORE_REF，禁止与 +3V3_INV 并联。
C69/C70/C71/C72/C76/C211 均 DNP；100 Ω 保留作串联隔离，常规滤波由 controller 实现。

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /ADC and status returns/
File: core_signal_protection.kicad_sch

Title: Baseboard — ADC and status return conditioning

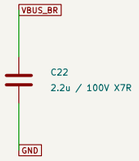
Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 17/19

母线电容 / 制动斩波 → VBUS → RS4（RS_BUS） 5 mΩ → VBUS_BR → 三相桥

RS4
5m / 3W / 1%





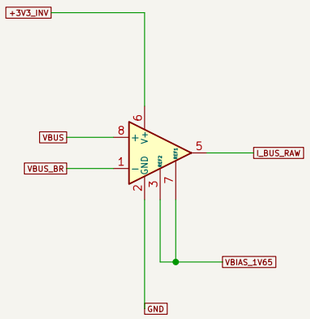




RS4（RS_BUS） 与 RS1-3 同型号/同封装：两端各独立取样线，四端开尔文走线交 #4。
C21/C22 共 4.4 µF 在桥侧：短路初期的桥侧放电不经分流，之后 C19/C20 放电经分流。

U33
INA241A2IDDFR





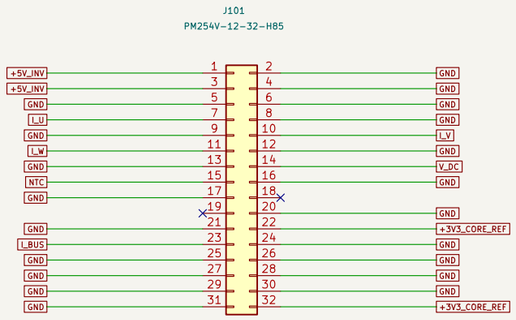


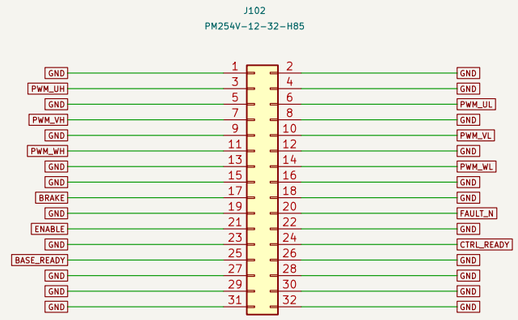


U33：与相电流同为 +3V3_INV：REF1/REF2 接同一 U13 缓冲的 VBIAS_1V65。
相电流无输入 RC，母线亦不另加输入 RC：100n 去耦、47 Ω 输出隔离沿用拓扑。
100 mV/A：零点 1.65 V：理想 ±16.5 A：最坏正向保证上限约 +14.5 A。

待用田决定：INA241 全温输出高端保证值 VS−0.2 V：3.3 V 只能保证 3.1 V，
达不到 +15 A 门限所需 3.15 V。A26 比较器/故障接入暂不落电路：不撞改 15 A。
两种改法：降低门限并留误差裕量，或改母线放大器供电（例如 +5V_INV，REF 不变）。
TLV7021 临时符号已准备：计算和极性处理见 issue3_changes.md。

power_system
Sheet: /母线电流（15 A保护待决）/
File: bus_current.kicad_sch
Title: 高端母线电流 / 保护供电待决
Size: A3 | Date: 2026-11-12 | Rev: E-1-issue3-draft
KiCad E.D.A. 10.0.7 | Id: 18/19





J101：1/3=+5V_INV；22/32=+3V3_CORE_REF；18/19 仅放 NC，禁止接地。
J101：20 个 GND；J102：21 个 GND。母座在 B 面（#4 翻到 B.Cu）。
与 controller 同号针一一对应，不做针号镜像或重排；同号焊盘 XY 由 #4 校核。







全部测试点放顶面（#4 落实）：ENABLE 为排针侧网络，板内保留 ENABLE_IN。

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system
Sheet: /64-pin core sockets/
File: power_interface.kicad_sch

Title: Baseboard minimum-core sockets

Size: A3	Date: 2026-11-12	**Rev: E-1-issue3-draft**
KiCad E.D.A. 10.0.7		Id: 20/19